Atlas Theory Integration Note

Summary. This new version is NOT trying to be a full theory of practical RLVR. Instead, it is a mathematically clean diagnostic suite that does four jobs: (i) it gives a correct finite-horizon perturbation statement for near-isospectrality; (ii) it gives a correct gap-based stability statement for dominant singular subspaces; (iii) it gives an exact checkpoint-level comparison between source-subspace models, one-sided hybrids, and target-subspace transport; and (iv) it gives a local geometric rationale for why a fixed-spectrum optimizer such as ISO is a coherent design choice.

I believe that is the right scope for an algorithmic paper whose main contribution is empirical and optimizer-driven, not a pure DL theory paper. The old asymptotic theorem is gone. The exact-exponential main theorem is gone. The ambient-transport theorem is now a checkpoint-level deterministic comparison theorem rather than a stylized theorem of RLVR dynamics. The truncation boundary issue is handled explicitly through an admissible-rank assumption. The gauge issue is handled at the class level rather than by privileging one chosen SVD basis. The one place where the paper can still be attacked is no longer theorem correctness, but theorem-to-paper mismatch if the manuscript does not update Section 4.2, the top-level prose, and the old appendices.

Scope and RLVR specificity. The results below should NOT be read as a derivation of near-isospectrality from the RLVR objective itself. As mathematical statements, the finite-horizon perturbation bounds, Wedin-type projector estimates, checkpoint reconstruction identities, and fixed-spectrum tangent characterization apply to arbitrary matrix paths or checkpoint pairs. **Their RLVR-specific content comes from the empirical regime documented in the paper**: in the studied reasoning-oriented RLVR runs, singular spectra change little relative to SFT, replacing the RL-trained spectrum by the base spectrum causes little performance loss, training only the spectrum does not yield comparable RLVR improvement, and checkpoint transitions are better fit by target-subspace transport than by source-subspace fitting.

Conditional on this empirical isospectral-transport regime, the theory supplies the appropriate gauge-invariant diagnostics for deciding whether the observed motion is source-subspace fitting, one-sided transport, or target-subspace transport. It also explains why ISO is a coherent optimizer design: ISO freezes the empirically stable spectral degrees of freedom and updates the Stiefel factors that carry the measured checkpoint motion. Thus, the role of the theory is to formalize, test, and exploit an observed RLVR geometry.

What it still does not do, and should not pretend to do, is prove that RLVR must be near-isospectral, or that ISO must beat AdamW or Muon. We shall clearly set out these in a limitation paragraph.

Are there still caveats reviewers can reasonably attack? Yes, but narrow rather than fatal.

1. The theory is conditional on a fixed informative truncation and a nontrivial boundary gap at the chosen rank. This is mathematically appropriate. The paper should therefore report the chosen rank and basic retained-energy and local-gap diagnostics.

2. The theory still does not prove that RLVR *must* be near-isospectral. Near-isospectrality remains an empirical finding of the paper.

3. The theory still does not prove that ISO must beat AdamW or Muon. Optimizer superiority remains empirical and should be carried by the trace-constraint ablation, time-to-target results, and tuned-baseline comparisons.

4. The checkpoint-comparison theorem is diagnostic and retrospective, not predictive. It explains which geometric model class best fits an observed transition between two checkpoints; it does not derive that transition from the RL objective.

These caveats are acceptable if we state them honestly and explicitly.

Does the theory now support the main paper backbone? Yes, but selectively.

1. It strongly supports the descriptive geometry story in Sections 3 and 4: finite-horizon spectral stability, gap-based subspace stability, and checkpoint-level preference for target-subspace transport over source-subspace models.

2. It supports ISO only as a *geometry-aligned design*. The local tangent-space proposition below explains why a fixed-spectrum parameterization removes explicit first-order spectrum-changing coordinates. **It does not prove that removing them is always beneficial**.

3. **It also does not support any claim that ISO is universally superior, universally more efficient, or theoretically optimal**. Those claims must remain empirical and narrow.

Is this theory novel at all? Our formal results use classical matrix perturbation and projection tools. The contribution is not a new perturbation theorem, but a diagnostic framework tailored to the empirical RLVR regime studied here: the results identify which quantities should be measured, which source-subspace baselines are appropriate, and why a fixed-spectrum optimizer is geometrically coherent once near-isospectral transport is observed. The paper-specific novelty to me is outlined as:

1. The exact model-class comparison between best source-subspace fit, one-sided hybrid fits, and target-subspace transport is paper-specific and directly aligned with the checkpoint analysis.

2. The theorem-matched evaluation protocol for Section 4.2 is a genuinely useful sharpening over the previous Procrustes-style story.

3. The local tangent-spectrum proposition gives a cleaner optimizer rationale than the old informal "Adam wastes off-manifold steps" story.

So the right positioning is: *classical tools, new assembly, directly matched to the empirical geometry and the optimizer design.* That is honest and still interesting. **BUT DO NOT OVER-HIGHLIGHT IT!**

1 How the theory should integrate back into the paper

Positioning sentence. The paper should use something close to the following.

Our main claims are empirical. The theory below is a diagnostic geometric lens: it explains small finite-horizon matrix drift, together with a nontrivial boundary gap, controls spectral drift and leading singular projectors, and it formalizes checkpoint-level tests that separate source-subspace fitting from target-subspace transport on truncated singular subspaces. It is not a literal convergence theory for noisy large-scale RLVR training, nor a theorem that ISO must outperform unconstrained optimizers.

Major claims, revised for the full paper storyline. These are the claims I would be willing to stand behind after the rewrite.

1. **Empirical geometry claim.** In the studied reasoning-oriented RLVR regimes, key weight matrices exhibit small finite-horizon spectral drift relative to SFT, and the dominant low-rank singular subspaces move nontrivially.

2. **Checkpoint-level diagnostic claim.** On an admissible fixed truncation, observed checkpoint transitions are better explained by target-subspace transport than by best source-subspace fitting; one-sided hybrids provide a finer diagnostic of whether the motion is mostly left, mostly right, or fully two-sided.

3. **Algorithmic claim.** Motivated by this geometry, ISO is a coherent fixed-spectrum optimizer design, and in the studied RLVR settings it improves stability and time-to-target while remaining competitive in final accuracy.

Claims the paper should stop making.

1. Do not say RLVR "honors the Stiefel manifold."

2. Do not say the theory proves RLVR is ambient transport.

3. Do not say the theory proves that AdamW wastes noise directions or that ISO must be better.

4. Do not say the paper establishes the same geometry for all RLHF or all larger-scale settings.

How theory and experiments should divide the labor.

Paper component	What theory certifies	What must remain empirical
Near-isospectrality in Section 3	If cumulative matrix drift is small over the observed horizon, spectral drift stays small; if a nontrivial boundary gap stays open, dominant projectors remain stable.	Whether RLVR actually lies in this regime. This comes from the measured spectra, Sigma-replacement, and Sigma-only ablations.
Section 4.2 geometry study	Exact comparison among best source-subspace fit, left-only and right-only hybrids, and target-subspace transport on truncated checkpoints.	Which model class wins on the actual checkpoints. This requires theorem-matched baselines in the experiment.
ISO design in Section 5	Fixed-spectrum updates remove explicit first-order spectrum-changing coordinates and stay on a coherent manifold of constant singular values.	Whether that removal improves stability or time-to-target. This comes from the trace-constraint ablation and optimizer comparisons.
Efficiency and accuracy claims in Section 6	Nothing.	Time-to-target, wall-clock tradeoff, tuned-baseline gap, stability at scale.

2 Finite-horizon spectral stability and subspace stability

Setup and notation. Let $W_t \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ denote the weight matrix of a fixed layer at RL step t, and let $q := \min(d_{\text{out}}, d_{\text{in}})$. Write

$$\sigma(W_t) := (\sigma_1(W_t), \ldots, \sigma_q(W_t)) \in \mathbb{R}^q_{\geq 0}$$

for the full singular-value vector in nonincreasing order, padded with zeros if necessary. For a fixed truncation rank $r < q$, let

$$W_t^{(r)} = U_t \Sigma_t V_t^\top, \qquad U_t \in \mathrm{St}(d_{\text{out}}, r),\ V_t \in \mathrm{St}(d_{\text{in}}, r),\ \Sigma_t = \mathrm{diag}(\sigma_1(W_t), \ldots, \sigma_r(W_t)).$$

Define the associated left and right projectors

$$P_t := U_t U_t^\top, \qquad Q_t := V_t V_t^\top.$$

All projector-based claims below are made on a fixed truncation rank r.

Definition 2.1 (Admissible truncation). *A fixed rank $r < q$ is called* admissible *for a checkpoint W if*

$$\sigma_r(W) > \sigma_{r+1}(W).$$

Then the leading rank-r singular subspaces, and hence the projectors P and Q, are canonically defined.

Remark 2.2 (Optional cluster variant)**.** *If the paper chooses an isolated singular-value cluster rather than a raw top-r cutoff, the projector statements below extend with the cluster projectors after replacing "top-r" by "chosen isolated cluster of size r." For maximal rigor and simplicity, the main text should avoid this extension unless it is actually needed. The cleanest paper is one that uses only empirically admissible top-r truncations.*

Theorem 2.3 (Finite-horizon spectral drift bound)**.** *Let $W_0, \dots, W_T \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and define $E_t := W_{t+1} - W_t$. Then for every $t \in \{1, \dots, T\}$,*

$$\|\sigma(W_t) - \sigma(W_0)\|_2 \le \|W_t - W_0\|_F \le \sum_{\tau=0}^{t-1} \|E_\tau\|_F \,, \tag{1}$$

$$\|\sigma(W_t) - \sigma(W_0)\|_\infty \le \|W_t - W_0\|_2 \le \sum_{\tau=0}^{t-1} \|E_\tau\|_2 \,. \tag{2}$$

Equivalently,

$$\frac{\|\sigma(W_t) - \sigma(W_0)\|_2}{\|\sigma(W_0)\|_2} \le \frac{\sum_{\tau=0}^{t-1} \|E_\tau\|_F}{\|W_0\|_F}.$$

For any fixed admissible rank-r truncation,

$$\|\Sigma_t - \Sigma_0\|_F \le \|\sigma(W_t) - \sigma(W_0)\|_2 \,.$$

Proof. The Hoffman-Wielandt-Mirsky inequality gives

$$\|\sigma(A) - \sigma(B)\|_2 \le \|A - B\|_F$$

for any two matrices A, B of the same size. Applying this with $A = W_t$ and $B = W_0$ yields the first inequality in (1). The second inequality in (1) follows from the telescoping identity

$$W_t - W_0 = \sum_{\tau=0}^{t-1} E_\tau$$

and the triangle inequality for the Frobenius norm.

Likewise, Weyl's singular-value perturbation bound gives

$$\|\sigma(A) - \sigma(B)\|_\infty \le \|A - B\|_2 \,.$$

Applying it to $(A, B) = (W_t, W_0)$ proves the first inequality in (2), and the second follows from the same telescoping identity plus the triangle inequality for the operator norm.

Finally, $\|\sigma(W_0)\|_2 = \|W_0\|_F$. The truncation bound follows because the diagonal entries of $\Sigma_t - \Sigma_0$ are a subset of the entries of $\sigma(W_t) - \sigma(W_0)$. □

Corollary 2.4 (Gap persistence under finite-horizon near-isospectrality)**.** *Fix $r < q$, and assume the rank-r truncation is admissible at W_0, with*

$$\delta_r(W_0) := \sigma_r(W_0) - \sigma_{r+1}(W_0) > 0.$$

If for all $t \le T$,

$$\sum_{\tau=0}^{t-1} \|E_\tau\|_2 \le \frac{\delta_r(W_0)}{4},$$

then for all $t \le T$,

$$\delta_r(W_t) := \sigma_r(W_t) - \sigma_{r+1}(W_t) \ge \frac{\delta_r(W_0)}{2}.$$

In particular, the same rank r remains admissible throughout the horizon.

Proof. By (2), each singular value moves by at most

$$\Delta_t := \sum_{\tau=0}^{t-1} \|E_\tau\|_2 .$$

Hence

$$\sigma_r(W_t) \geq \sigma_r(W_0) - \Delta_t, \qquad \sigma_{r+1}(W_t) \leq \sigma_{r+1}(W_0) + \Delta_t.$$

Subtracting gives

$$\delta_r(W_t) \geq \delta_r(W_0) - 2\Delta_t.$$

If $\Delta_t \leq \delta_r(W_0)/4$, then $\delta_r(W_t) \geq \delta_r(W_0)/2$. $\square$

Wedin-type one-step projector bound. Let $A, \widetilde{A} = A + E \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$. Let U, V and $\widetilde{U}, \widetilde{V}$ denote orthonormal bases for the leading rank-r left and right singular subspaces of A and $\widetilde{A}$, respectively. Assume the two leading-block cross-gaps

$$\gamma_1 := \sigma_r(A) - \sigma_{r+1}(\widetilde{A}) > 0, \qquad \gamma_2 := \sigma_r(\widetilde{A}) - \sigma_{r+1}(A) > 0,$$

and set $\gamma := \min\{\gamma_1, \gamma_2\}$. Then Wedin's $\sin\Theta$ theorem implies

$$\max\{\left\|\sin\Theta(U, \widetilde{U})\right\|_2, \left\|\sin\Theta(V, \widetilde{V})\right\|_2\} \leq C_{\text{Wed}} \frac{\|E\|_2}{\gamma},$$

where C_{Wed} is a universal numerical constant. This is the only form of Wedin's theorem used below; no sharp constant is needed for the paper's claims.

Theorem 2.5 (Gap-based stability of the leading singular subspace)**.** *Assume the rank-r truncation is admissible for every checkpoint $W_0, \ldots, W_T$. Fix $r < q$. For each $t = 0, \ldots, T-1$, let*

$$\delta_t := \sigma_r(W_t) - \sigma_{r+1}(W_t) > 0, \qquad E_t := W_{t+1} - W_t.$$

Assume $\|E_t\|_2 \leq \delta_t/2$ for every t. Let C_{Wed} denote the universal constant in the Wedin-type bound above. Let

$$P_t := U_t U_t^\top, \qquad Q_t := V_t V_t^\top$$

be the projectors onto the leading rank-r left and right singular subspaces of W_t. Then

$$\|P_{t+1} - P_t\|_2 \leq \frac{2C_{\text{Wed}} \|E_t\|_2}{\delta_t}, \tag{3}$$

$$\|Q_{t+1} - Q_t\|_2 \leq \frac{2C_{\text{Wed}} \|E_t\|_2}{\delta_t}. \tag{4}$$

Consequently,

$$\|P_T - P_0\|_2 \leq \sum_{t=0}^{T-1} \frac{2C_{\text{Wed}} \|E_t\|_2}{\delta_t}, \tag{5}$$

$$\|Q_T - Q_0\|_2 \leq \sum_{t=0}^{T-1} \frac{2C_{\text{Wed}} \|E_t\|_2}{\delta_t}. \tag{6}$$

In particular, if $\delta_t \geq \underline{\delta} > 0$ for all $t \leq T-1$, then

$$\|P_T - P_0\|_2, \ \|Q_T - Q_0\|_2 \leq \frac{2C_{\text{Wed}}}{\underline{\delta}} \sum_{t=0}^{T-1} \|E_t\|_2 .$$

Proof. Fix t. By Weyl's inequality,

$$\sigma_r(W_{t+1}) \geq \sigma_r(W_t) - \|E_t\|_2 , \qquad \sigma_{r+1}(W_{t+1}) \leq \sigma_{r+1}(W_t) + \|E_t\|_2 .$$

Therefore both cross-gaps

$$\sigma_r(W_t) - \sigma_{r+1}(W_{t+1}), \qquad \sigma_r(W_{t+1}) - \sigma_{r+1}(W_t)$$

are at least

$$\gamma_t := \delta_t - \|E_t\|_2 \geq \frac{\delta_t}{2}.$$

Applying the Wedin-type one-step projector bound above to the leading rank-r left and right singular subspaces of W_t and W_{t+1} yields

$$\max\{\|\sin\Theta(U_t, U_{t+1})\|_2 , \|\sin\Theta(V_t, V_{t+1})\|_2\} \leq C_{\mathrm{Wed}} \frac{\|E_t\|_2}{\gamma_t} \leq \frac{2C_{\mathrm{Wed}} \|E_t\|_2}{\delta_t}.$$

For orthogonal projectors, the standard identity

$$\|P_{t+1} - P_t\|_2 = \|\sin\Theta(U_t, U_{t+1})\|_2$$

gives (3). The displayed maximum bound gives (4) as well. Summing the one-step bounds and using the triangle inequality yields (5) and (6). □

This result controls the conditioning and trackability of the retained projectors. But it does not assert that the projectors are fixed; nonzero projector motion is precisely what the checkpoint diagnostics in Section 3 measure.

Corollary 2.6 (Projector stability from an initial gap)**.** *Fix $r < q$, let $\delta_r(W_0) > 0$, and assume*

$$\sum_{\tau=0}^{t-1} \|E_\tau\|_2 \leq \frac{\delta_r(W_0)}{4} \qquad \textit{for all } t \leq T.$$

Then

$$\|P_T - P_0\|_2 , \|Q_T - Q_0\|_2 \leq \frac{4C_{\mathrm{Wed}}}{\delta_r(W_0)} \sum_{t=0}^{T-1} \|E_t\|_2 .$$

Proof. By Corollary 2.4, the rank-r gap stays at least $\delta_r(W_0)/2$ throughout the horizon. Also, for each step,

$$\|E_t\|_2 \leq \sum_{\tau=0}^{t} \|E_\tau\|_2 \leq \frac{\delta_r(W_0)}{4} \leq \frac{\delta_t}{2},$$

so the one-step hypothesis of Theorem 2.5 holds throughout the horizon. Applying that theorem with $\underline{\delta} = \delta_r(W_0)/2$ yields the claim. □

Remark 2.7 (What Section 3 of the paper should claim)**.** *Theorems 2.3 and 2.5 do not say that RLVR is generically isospectral. They say that if the empirical training path has small finite-horizon matrix drift, singular values remain stable; if, in addition, a nontrivial rank-r boundary gap persists, dominant projectors remain stable. That is the right way to support Section 3.*

3 Checkpoint-level geometric diagnosis on truncated singular subspaces

Theorem 3.1 (Exact checkpoint-level comparison of geometric model classes)**.** *Fix an admissible truncation rank $r < q$ and consider two checkpoints*

$$W_j^{(r)} = U_j \Sigma_j V_j^\top, \qquad W_i^{(r)} = U_i \Sigma_i V_i^\top,$$

with $U_j, U_i \in \mathrm{St}(d_{\mathrm{out}}, r)$, $V_j, V_i \in \mathrm{St}(d_{\mathrm{in}}, r)$, and diagonal $\Sigma_j, \Sigma_i \in \mathbb{R}^{r \times r}$ with nonincreasing nonnegative diagonals. Let

$$P_j := U_j U_j^\top, \quad Q_j := V_j V_j^\top, \quad P_i := U_i U_i^\top, \quad Q_i := V_i V_i^\top.$$

Define the following gauge-invariant model classes:

$$\begin{aligned}
\mathcal{I}(j) &:= \{A : P_j A = A = A Q_j\} = \{U_j M V_j^\top : M \in \mathbb{R}^{r \times r}\}, \\
\mathcal{L}(j \to i) &:= \{A : P_j A = A = A Q_i\} = \{U_j M V_i^\top : M \in \mathbb{R}^{r \times r}\}, \\
\mathcal{R}(j \to i) &:= \{A : P_i A = A = A Q_j\} = \{U_i M V_j^\top : M \in \mathbb{R}^{r \times r}\}, \\
\mathcal{A}(j \to i) &:= \{A : A = U \Sigma_j V^\top,\ UU^\top = P_i,\ VV^\top = Q_i,\ U^\top U = V^\top V = I_r\}.
\end{aligned}$$

Then the following hold.

(i) Exact inner mixing preserves projectors. *If there exist $R_U, R_V \in O(r)$ such that*

$$U_i = U_j R_U, \qquad V_i = V_j R_V,$$

then $P_i = P_j$ and $Q_i = Q_j$. Hence any nontrivial projector change is incompatible with exact inner mixing.

(ii) Best source-subspace fit. *The orthogonal projection of an arbitrary matrix A onto $\mathcal{I}(j)$ is $P_j A Q_j$. In particular,*

$$\inf_{B \in \mathcal{I}(j)} \left\| W_i^{(r)} - B \right\|_F = \left\| W_i^{(r)} - P_j W_i^{(r)} Q_j \right\|_F, \tag{7}$$

and

$$\left\| W_i^{(r)} - P_j W_i^{(r)} Q_j \right\|_F^2 = \left\| (I - P_j) W_i^{(r)} \right\|_F^2 + \left\| P_j W_i^{(r)} (I - Q_j) \right\|_F^2. \tag{8}$$

(iii) Best one-sided hybrids.

$$\inf_{B \in \mathcal{L}(j \to i)} \left\| W_i^{(r)} - B \right\|_F = \left\| (I - P_j) W_i^{(r)} \right\|_F, \qquad \arg\min_{B \in \mathcal{L}(j \to i)} \left\| W_i^{(r)} - B \right\|_F = P_j W_i^{(r)}, \tag{9}$$

$$\inf_{B \in \mathcal{R}(j \to i)} \left\| W_i^{(r)} - B \right\|_F = \left\| W_i^{(r)} (I - Q_j) \right\|_F, \qquad \arg\min_{B \in \mathcal{R}(j \to i)} \left\| W_i^{(r)} - B \right\|_F = W_i^{(r)} Q_j. \tag{10}$$

(iv) Best target-subspace transport fit. *The class $\mathcal{A}(j \to i)$ depends only on the target projectors (P_i, Q_i) and the source singular values Σ_j, not on any chosen SVD gauge at checkpoint i. Its best approximation error is exactly*

$$\inf_{A \in \mathcal{A}(j \to i)} \left\| W_i^{(r)} - A \right\|_F = \left\| \Sigma_i - \Sigma_j \right\|_F. \tag{11}$$

One minimizer is given by

$$\widehat{W}_{j \to i}^{\mathrm{amb}} := U_i \Sigma_j V_i^\top.$$

*(v) **Exact model-class comparison.*** *Ambient transport explains the checkpoint transition better than every source-subspace model in the fixed source-subspace class $\mathcal{I}(j)$ if and only if*

$$\|\Sigma_i - \Sigma_j\|_F < \left\|W_i^{(r)} - P_j W_i^{(r)} Q_j\right\|_F. \tag{12}$$

Likewise, the one-sided comparisons are exactly

$$\mathcal{A}(j \to i) \text{ beats } \mathcal{L}(j \to i) \iff \|\Sigma_i - \Sigma_j\|_F < \left\|(I - P_j)W_i^{(r)}\right\|_F, \tag{13}$$

$$\mathcal{A}(j \to i) \text{ beats } \mathcal{R}(j \to i) \iff \|\Sigma_i - \Sigma_j\|_F < \left\|W_i^{(r)}(I - Q_j)\right\|_F. \tag{14}$$

Proof. **Part (i).** If $U_i = U_j R_U$ with $R_U^\top R_U = I$, then

$$P_i = U_i U_i^\top = U_j R_U R_U^\top U_j^\top = U_j U_j^\top = P_j.$$

The proof for $Q_i = Q_j$ is identical.

Part (ii). The set $\mathcal{I}(j)$ is a linear subspace of $\mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$. We claim that the orthogonal projector onto this subspace, with respect to the Frobenius inner product, is

$$\Pi_{\mathcal{I}(j)}(A) = P_j A Q_j.$$

Indeed, $P_j A Q_j \in \mathcal{I}(j)$ because

$$P_j A Q_j = U_j (U_j^\top A V_j) V_j^\top.$$

For any $B = U_j M V_j^\top \in \mathcal{I}(j)$,

$$\langle A - P_j A Q_j, B\rangle_F = \operatorname{tr}\big((A - P_j A Q_j)^\top U_j M V_j^\top\big) = \operatorname{tr}\big(V_j^\top (A - P_j A Q_j)^\top U_j M\big).$$

But

$$U_j^\top (A - P_j A Q_j) V_j = U_j^\top A V_j - U_j^\top P_j A Q_j V_j = U_j^\top A V_j - U_j^\top A V_j = 0,$$

so the Frobenius inner product is zero. Hence $\Pi_{\mathcal{I}(j)}(A) = P_j A Q_j$. Applying this to $A = W_i^{(r)}$ proves (7).

To prove (8), write

$$W_i^{(r)} - P_j W_i^{(r)} Q_j = (I - P_j) W_i^{(r)} + P_j W_i^{(r)} (I - Q_j).$$

The two summands are Frobenius-orthogonal because

$$\left\langle (I - P_j) W_i^{(r)}, P_j W_i^{(r)} (I - Q_j)\right\rangle_F = \operatorname{tr}\big((W_i^{(r)})^\top (I - P_j) P_j W_i^{(r)} (I - Q_j)\big) = 0.$$

Taking squared Frobenius norms gives (8).

Part (iii). The orthogonal projector onto $\mathcal{L}(j \to i)$ is $A \mapsto P_j A Q_i$. Since $W_i^{(r)} Q_i = W_i^{(r)}$, the best approximation to $W_i^{(r)}$ in $\mathcal{L}(j \to i)$ is $P_j W_i^{(r)}$. The residual is

$$W_i^{(r)} - P_j W_i^{(r)} = (I - P_j) W_i^{(r)},$$

which proves the left identity in (9). The right identity is symmetric, since the orthogonal projector onto $\mathcal{R}(j \to i)$ is $A \mapsto P_i A Q_j$, and $P_i W_i^{(r)} = W_i^{(r)}$.

Part (iv). Fix any orthonormal bases U_i, V_i of the target subspaces. Every element of $\mathcal{A}(j \to i)$ can be written as

$$A = U_i R_L \Sigma_j R_R^\top V_i^\top \qquad \text{for some } R_L, R_R \in O(r).$$

Therefore

$$\inf_{A\in\mathcal{A}(j\to i)} \left\|W_i^{(r)} - A\right\|_F = \inf_{R_L,R_R\in O(r)} \left\|\Sigma_i - R_L\Sigma_j R_R^\top\right\|_F,$$

using Frobenius-norm invariance under left and right multiplication by matrices with orthonormal columns. Now

$$\left\|\Sigma_i - R_L\Sigma_j R_R^\top\right\|_F^2 = \|\Sigma_i\|_F^2 + \|\Sigma_j\|_F^2 - 2\mathrm{tr}\big(\Sigma_i R_L\Sigma_j R_R^\top\big).$$

By von Neumann's trace inequality,

$$\mathrm{tr}\big(\Sigma_i R_L\Sigma_j R_R^\top\big) \le \sum_{k=1}^{r} \sigma_k(\Sigma_i)\sigma_k(\Sigma_j) = \mathrm{tr}(\Sigma_i\Sigma_j),$$

where the last equality holds because Σ_i and Σ_j are diagonal with nonnegative entries in nonincreasing order. Equality is attained by $R_L = R_R = I$. Hence

$$\inf_{R_L,R_R\in O(r)} \left\|\Sigma_i - R_L\Sigma_j R_R^\top\right\|_F = \|\Sigma_i - \Sigma_j\|_F,$$

which proves (11). $U_i\Sigma_j V_i^\top$ is therefore a minimizer.

Part (v). All three comparison statements follow immediately from parts (ii)–(iv). □

Corollary 3.2 (Principal-angle certificate)**.** *Under the setup of Theorem 3.1,*

$$\inf_{B\in\mathcal{I}(j)} \left\|W_i^{(r)} - B\right\|_F \ge \max\Big\{\left\|(I-P_j)W_i^{(r)}\right\|_F,\ \left\|W_i^{(r)}(I-Q_j)\right\|_F\Big\} \tag{15}$$

$$\ge \sigma_r(W_i)\max\Big\{\|\sin\Theta(U_i,U_j)\|_F,\ \|\sin\Theta(V_i,V_j)\|_F\Big\}. \tag{16}$$

Consequently, a sufficient certificate for ambient transport to beat every model in the fixed source-subspace class $\mathcal{I}(j)$ is

$$\|\Sigma_i - \Sigma_j\|_F < \sigma_r(W_i)\max\Big\{\|\sin\Theta(U_i,U_j)\|_F,\ \|\sin\Theta(V_i,V_j)\|_F\Big\}. \tag{17}$$

If one wants a sufficient certificate for target-subspace transport to beat both one-sided hybrid classes $\mathcal{L}(j\to i)$ and $\mathcal{R}(j\to i)$, a valid stronger condition is

$$\|\Sigma_i - \Sigma_j\|_F < \sigma_r(W_i)\min\Big\{\|\sin\Theta(U_i,U_j)\|_F,\ \|\sin\Theta(V_i,V_j)\|_F\Big\}. \tag{18}$$

Proof. For any $B\in\mathcal{I}(j)$, we have $(I-P_j)B = 0$ and $B(I-Q_j) = 0$. Hence

$$(I-P_j)(W_i^{(r)} - B) = (I-P_j)W_i^{(r)}, \qquad (W_i^{(r)} - B)(I-Q_j) = W_i^{(r)}(I-Q_j).$$

Contractivity of orthogonal projectors in Frobenius norm gives

$$\left\|W_i^{(r)} - B\right\|_F \ge \left\|(I-P_j)W_i^{(r)}\right\|_F, \qquad \left\|W_i^{(r)} - B\right\|_F \ge \left\|W_i^{(r)}(I-Q_j)\right\|_F.$$

Taking the infimum over $B\in\mathcal{I}(j)$ proves (15).

Next,

$$\left\|(I-P_j)W_i^{(r)}\right\|_F = \left\|(I-P_j)U_i\Sigma_i V_i^\top\right\|_F = \|(I-P_j)U_i\Sigma_i\|_F \ge \sigma_r(W_i)\,\|(I-P_j)U_i\|_F.$$

The singular values of $(I-P_j)U_i$ are the sines of the principal angles between $\mathrm{range}(U_i)$ and $\mathrm{range}(U_j)$, so

$$\|(I-P_j)U_i\|_F = \|\sin\Theta(U_i,U_j)\|_F .$$

Thus

$$\left\|(I-P_j)W_i^{(r)}\right\|_F \geq \sigma_r(W_i)\,\|\sin\Theta(U_i,U_j)\|_F .$$

The right-subspace estimate is identical:

$$\left\|W_i^{(r)}(I-Q_j)\right\|_F = \left\|U_i\Sigma_iV_i^\top(I-Q_j)\right\|_F = \left\|\Sigma_iV_i^\top(I-Q_j)\right\|_F \geq \sigma_r(W_i)\,\|\sin\Theta(V_i,V_j)\|_F .$$

Combining these inequalities with (15) proves (16). The final certificate follows by comparing (11) with (16). The one-sided hybrid certificate follows by comparing (11) with (13) and (14), together with the two one-sided principal-angle lower bounds established above. □

Corollary 3.3 (Gauge-invariant evaluation protocol for Section 4.2)**.** *Under the setup of Theorem 3.1, define*

$$e^\star_{\mathrm{src}} := \frac{\inf_{B\in\mathcal{I}(j)}\left\|W_i^{(r)}-B\right\|_F}{\left\|W_i^{(r)}\right\|_F}, \qquad e^\star_{\mathrm{L}} := \frac{\inf_{B\in\mathcal{L}(j\to i)}\left\|W_i^{(r)}-B\right\|_F}{\left\|W_i^{(r)}\right\|_F},$$

$$e^\star_{\mathrm{R}} := \frac{\inf_{B\in\mathcal{R}(j\to i)}\left\|W_i^{(r)}-B\right\|_F}{\left\|W_i^{(r)}\right\|_F}, \quad e^\star_{\mathrm{amb}} := \frac{\inf_{A\in\mathcal{A}(j\to i)}\left\|W_i^{(r)}-A\right\|_F}{\left\|W_i^{(r)}\right\|_F}.$$

Then

$$e^\star_{\mathrm{src}} = \frac{\left\|W_i^{(r)}-P_jW_i^{(r)}Q_j\right\|_F}{\|\Sigma_i\|_F}, \qquad e^\star_{\mathrm{L}} = \frac{\left\|(I-P_j)W_i^{(r)}\right\|_F}{\|\Sigma_i\|_F},$$

$$e^\star_{\mathrm{R}} = \frac{\left\|W_i^{(r)}(I-Q_j)\right\|_F}{\|\Sigma_i\|_F}, \qquad e^\star_{\mathrm{amb}} = \frac{\|\Sigma_i-\Sigma_j\|_F}{\|\Sigma_i\|_F}.$$

These four relative errors are invariant under admissible SVD gauge changes. The ambient error depends only on Σ_i and Σ_j, while the source and hybrid errors are computed from the truncated target operator $W_i^{(r)}$ and the relevant projectors. They are not claimed to be determined by the aggregate projectors and spectra alone.

Proof. The exact formulas follow immediately from Theorem 3.1 and the identity $\left\|W_i^{(r)}\right\|_F = \|\Sigma_i\|_F$. Each expression is an orthogonal-projection residual written in terms of $W_i^{(r)}$, the relevant projectors, and the retained spectra, and is therefore unchanged under admissible changes of SVD representatives. Because unequal singular values weight directions inside the retained subspace, the source and hybrid residuals generally depend on the actual truncated operator $W_i^{(r)}$, not only on aggregate projectors and spectra. □

Remark 3.4 (Why the test must be truncated)**.** *If one uses a full-rank square decomposition with $r=q=d_{\mathrm{out}}=d_{\mathrm{in}}$, then $P_t=I$ and $Q_t=I$ for every t, so projector motion becomes vacuous. The source-subspace versus target-subspace distinction is therefore meaningful only on a fixed informative truncation $r<q$.*

Remark 3.5 (What Section 4.2 should now claim)**.** *Theorem 3.1 does not prove that RLVR must be ambient transport. It proves a checkpoint-level model-selection statement: on an admissible fixed truncation, if the target-subspace transport class fits better than the source-subspace and hybrid classes, then the observed transition is geometrically better explained by transport than by staying inside the source subspaces. This is exactly the right strength for Section 4.2.*

4 Connection to ISO and the algorithmic motivation

Proposition 4.1 (First-order spectrum-changing coordinates and fixed-spectrum tangent directions)**.** *Assume*

$$W = U\Sigma V^\top, \qquad U \in \mathrm{St}(d_{\mathrm{out}}, r),\ V \in \mathrm{St}(d_{\mathrm{in}}, r),$$

where $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$ has positive simple *diagonal entries. Let*

$$\mathcal{M}_\Sigma := \{\widetilde{U}\Sigma\widetilde{V}^\top : \widetilde{U} \in \mathrm{St}(d_{\mathrm{out}}, r),\ \widetilde{V} \in \mathrm{St}(d_{\mathrm{in}}, r)\}$$

be the fixed-spectrum manifold through W. Because the retained singular values are assumed positive and simple, the following tangent statements are local at the displayed SVD representative, up to the usual finite sign gauge. With repeated singular values, the same set should be treated as a fixed-spectrum orbit with within-cluster gauge freedom.

(i) If $W(\varepsilon) \in \mathcal{M}_\Sigma$ is a differentiable curve with $W(0) = W$ and tangent $\dot{W} := \frac{d}{d\varepsilon}W(\varepsilon)|_{\varepsilon=0}$, then

$$\mathrm{diag}(U^\top \dot{W} V) = 0.$$

So every tangent direction to $\mathcal{M}_\Sigma$ has zero first-order singular-value drift in singular coordinates.

(ii) Let G be any perturbation matrix and consider $W(\varepsilon) = W + \varepsilon G$. Then, for each retained singular value,

$$\frac{d}{d\varepsilon}\sigma_k(W(\varepsilon))\bigg|_{\varepsilon=0} = u_k^\top G v_k, \qquad k = 1, \ldots, r.$$

Equivalently, the diagonal of $U^\top G V$ is exactly the vector of first-order singular-value changes.

(iii) If D is any diagonal matrix in $\mathbb{R}^{r\times r}$, then the perturbation $UDV^\top$ is Frobenius-orthogonal to every tangent direction of $\mathcal{M}_\Sigma$ at W.

Thus the diagonal core of $U^\top G V$ is the explicit first-order spectrum-changing component of a general Euclidean perturbation, and fixed-spectrum optimization removes it by construction.

Proof. For part (i), let $W(\varepsilon) = U(\varepsilon)\Sigma V(\varepsilon)^\top$ with $U(0) = U$ and $V(0) = V$. Since $U(\varepsilon)^\top W(\varepsilon) V(\varepsilon) = \Sigma$ for all ε, differentiating at 0 gives

$$\dot{U}^\top U \Sigma + U^\top \dot{W} V + \Sigma V^\top \dot{V} = 0.$$

Since $U(\varepsilon)$ and $V(\varepsilon)$ remain orthonormal, the matrices $\dot{U}^\top U$ and $V^\top \dot{V}$ are skew-symmetric. Taking diagonals in the displayed identity yields

$$\mathrm{diag}(U^\top \dot{W} V) = 0,$$

because the diagonal of a skew-symmetric matrix is zero and multiplying by a diagonal matrix preserves zero diagonal.

For part (ii), let $u_k(\varepsilon), v_k(\varepsilon)$ denote differentiable singular vectors associated with the simple singular value $\sigma_k(\varepsilon)$ of $W(\varepsilon) = W + \varepsilon G$. Since

$$\sigma_k(\varepsilon) = u_k(\varepsilon)^\top W(\varepsilon) v_k(\varepsilon),$$

differentiating at 0 gives

$$\sigma_k'(0) = \dot{u}_k^\top W v_k + u_k^\top G v_k + u_k^\top W \dot{v}_k.$$

Using $Wv_k = \sigma_k u_k$ and $u_k^\top W = \sigma_k v_k^\top$, this becomes

$$\sigma_k'(0) = \sigma_k \dot{u}_k^\top u_k + u_k^\top G v_k + \sigma_k v_k^\top \dot{v}_k.$$

Because $u_k(\varepsilon)$ and $v_k(\varepsilon)$ remain unit vectors, differentiating $u_k^\top u_k = 1$ and $v_k^\top v_k = 1$ shows $\dot{u}_k^\top u_k = 0$ and $v_k^\top \dot{v}_k = 0$. Therefore

$$\sigma_k'(0) = u_k^\top G v_k.$$

Collecting these identities for $k = 1, \dots, r$ yields the stated diagonal formula.

For part (iii), every tangent direction has the form

$$\dot{W} = \dot{U}\,\Sigma V^\top + U\Sigma \dot{V}^\top, \qquad \dot{U} \in T_U \mathrm{St}(d_{\mathrm{out}}, r),\ \dot{V} \in T_V \mathrm{St}(d_{\mathrm{in}}, r).$$

Now

$$\left\langle UDV^\top, \dot{U}\,\Sigma V^\top \right\rangle_F = \mathrm{tr}\big(D\,U^\top \dot{U}\,\Sigma\big), \qquad \left\langle UDV^\top, U\Sigma\dot{V}^\top \right\rangle_F = \mathrm{tr}\big(D\Sigma\,\dot{V}^\top V\big).$$

Both $U^\top \dot{U}$ and $\dot{V}^\top V$ are skew-symmetric, while $D\Sigma$ is diagonal, so both traces are zero. Therefore $\left\langle UDV^\top, \dot{W} \right\rangle_F = 0$. □

Remark 4.2 (How Proposition 4.1 should be used)**.** *Proposition 4.1 gives a clean* local *rationale for ISO: fixed-spectrum updates null out the explicit first-order singular-value drift coordinates. This supports ISO as a coherent geometry-aligned design. It does* not *prove that those coordinates are useless, noisy, or harmful in practical RLVR, and it does not prove that ISO should beat AdamW or Muon. That part of the story must stay empirical.*

Remark 4.3 (Why this proposition helps motivate ISO)**.** *This proposition lets the paper say something stronger and cleaner than "Adam explores noise." The defensible claim is:* relative to a general Euclidean update, a fixed-spectrum parameterization removes explicit first-order singular-value drift coordinates, and the paper's empirical results indicate that doing so is helpful in the studied RLVR regime. *That is honest and well aligned with the trace-constraint ablation.*

5 Paper-wide rewrite map

KEY checklists before submission

1. Rerun Section 4.2 with the theorem-matched baselines in Corollary 3.3.
2. Report the chosen truncation rank r, retained energy, and local boundary gap for representative analyzed layers.
3. Stop using vector-by-index singular-vector plots as formal evidence in the main text.
4. Put the trace-constraint ablation at the center of the optimizer motivation.
5. Keep the final-score narrative conservative against tuned AdamW, and present the main win as stability and time-to-target.
6. Delete or explicitly relabel the old optimizer appendix and remove the theoremized gauge appendix.

Abstract & contribution bullets. These should be softened and made theorem-compatible.

1. **TL;DR.** Replace "RL post-training for LLMs honors Stiefel manifold and enforce this geometry for more efficient and effective alignment" with something like:

   > We empirically find that RLVR in the studied reasoning settings is nearly isospectral and primarily moves dominant singular subspaces; ISO exploits this geometry and improves stability and time-to-target in these settings.

2. **Abstract.** Replace "RLVR is dominated by ambient transport" with "on the truncated checkpoint transitions we analyze, target-subspace transport explains the observed movement substantially better than source-subspace models."

3. **Contributions.** The second bullet should no longer say "ambient transport is the dominant mode" without qualification. It should say "we provide a checkpoint-level geometric diagnosis showing that, on admissible truncated subspaces, the observed transitions are better explained by target-subspace transport than by source-subspace models."

Section 2. The current "Take-away 2: Full-parameter AdamW is a suboptimal default" is too strong. Replace it with a milder statement such as: "These observations suggest that full-parameter AdamW is not obviously optimal for RLVR and that geometry-aware optimizer design is worth investigating."

Section 3.1. Delete the asymptotic notation $\Delta_\Sigma(t) = o(1)$ and any text claiming vanishing relative drift. Replace it by a finite-horizon empirical statement tied to Figure 2, and cite Theorem 2.3 only as the perturbation lens that justifies the language "small over the observed horizon."

Section 3.4. The raw singular-vector-by-index plots should not carry formal claims. Keep them only as qualitative visualizations or move them to the appendix. The formal story in the main text should move to projector distances, principal angles between retained subspaces, and the Section 4.2 reconstruction diagnostics.

Section 4.1. Replace the old exact-exponential theorem with Theorem 2.5. If you want to keep a mention of approximate exponential decay, demote it to a short intuition remark or appendix note, not a main theorem assumption.

Section 4.2. This is the single most important rewrite.

1. Rename Hypothesis A from "inner mixing" to "best source-subspace model".

2. Report the theorem-matched baselines from Corollary 3.3: source-subspace, left-only hybrid, right-only hybrid, and target-subspace transport.

3. Use

   $$\widehat{W}^{\mathrm{src}}_{j\to i} = P_j W_i^{(r)} Q_j, \qquad \widehat{W}^{\mathrm{L}}_{j\to i} = P_j W_i^{(r)}, \qquad \widehat{W}^{\mathrm{R}}_{j\to i} = W_i^{(r)} Q_j, \qquad \widehat{W}^{\mathrm{amb}}_{j\to i} = U_i \Sigma_j V_i^\top.$$

   The old Procrustes-only baseline can remain as a secondary visualization, but not as the main null.

4. State the truncation rank r in the main text and figure captions, and report retained energy and local boundary gap for representative checkpoints.

Section 5. Use Proposition 4.1 only as a local geometric motivation. The main optimizer paragraph should say that ISO removes first-order singular-value drift coordinates by construction, while the case that this helps RLVR is made empirically by the trace-constraint ablation, Sigma-only ablation, time-to-target comparison, and tuned-baseline results.

Section 6. The main performance narrative should be "better stability and time-to-target in the studied RLVR settings," not "theory-backed optimizer superiority." The tuned-Adam final-accuracy gap should be presented as modest. The 32B result should be framed as a large-scale stability result, not as a full engineering-efficiency proof.

Conclusion and limitations. Add an explicit limitations paragraph with the following points: (1) our theory is diagnostic, not predictive; (2) the geometry claims are conditional on informative truncation and admissible boundary gap; (3) the optimizer theory is only local and does not prove ISO must outperform unconstrained baselines; (4) the current evidence is strongest for reasoning-oriented RLVR with verifiable or strongly structured rewards.

Appendices. Delete the current optimizer convergence appendix, unless it is rewritten to be explicitly about an idealized method rather than Algorithm 1. Demote the current SVD-gauge appendix to a short remark rather than a theorem section.